Turbine Temperature Indicating System – General Data

1. General

The turbine temperature (T_5) indicating system provides an accurate indication of engine operating temperatures taken at a point between the two turbines. The system consists essentially of a thermocouple assembly, mounted around the power turbine stator housing, and a turbine temperature indication displayed in the engine parameters display provided in the upper inboard 1/6th portion of the primary flight display.

A. Turbine Temperature Indication

Turbine temperature (T_5) is displayed with a pointer type display with integer values beneath each display. Units are displayed in °C.

B. Thermocouple Assembly

The thermocouple assembly consists of 8 individual chromel-alumel thermocouple probes connected in parallel, two bus-bars, twin leads and a connector. Each probe protrudes through a threaded boss into the turbine stator housing between the two turbines and may be removed and replaced individually.

2. Adjustment/Test

A. Calibration Check

CAUTION **EXERCISE CARE WHEN REMOVING OR INSTALLING TERMINAL BOLTS. EXCESSIVE PRESSURE ON LUGS COULD RESULT IN DAMAGE TO INSULATION.**

Equipment required:

- Calibrated thermocouple source
- Multimeter – Avo 8 (or equivalent)

1) Disconnect turbine temperature leads from thermocouple harness terminals at engine.

2) Using a multimeter, check insulation and resistance limits of T_5 thermocouple harness at engine thermocouple harness terminals.

- Insulation (terminals to ground) – not less than 50,000 ohms
- Resistance (between terminals) – between 0.58 and 0.74 ohm

3) Using the thermocouple source, connect to the disconnected thermocouple leads on the aircraft side.

a) Set thermocouple source output to 0°C (0µV) and record the output temperature displayed on Apex.

b) Set thermocouple source output to 600°C (24901.6µV) and record the output temperature displayed on Apex.

c) Set thermocouple source output to 700°C (29128.3µV) and record the output temperature displayed on Apex.

d) Set thermocouple source output to 800°C (33276.8µV) and record the output temperature displayed on Apex.

Ensure readings are within the following accuracies:

- –55°C to 300°C ± 36°C
- 300°C to 800°C ± 15°C
- 700°C to 800°C ± 10°C
- 800°C to 1200°C –5°C/+10°C

4) Reconnect turbine temperature leads to thermocouple harness terminals at engine.

Print Date: 2025-09-18